GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

Question Bank 5.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the copy's perimeter or area.

1. Perimeter 22 cm, k = 2.

44 cm

2. Area 5 cm^2, k = 4.

80 cm^2

5 × 16.

3. Area 72 cm^2, k = ⅓.

8 cm^2

72 × $\frac{1}{9}$.

4. A 3 cm by 7 cm rectangle, k = 2. Perimeter and area of the copy.

P = 40 cm, A = 84 cm^2

20 × 2; 21 × 4.

For questions 5 through 7, find k, or the ratio asked.

5. Areas 6 cm^2 and 150 cm^2. Find k.

k = 5

150 ÷ 6 = 25 = k^2.

6. Areas 50 cm^2 and 8 cm^2, original to copy. Find k.

k = 0.4

8 ÷ 50 = 0.16 = k^2, so k = 0.4: the copy is smaller.

7. Perimeters 15 cm and 45 cm, original to copy. What is the area ratio?

9

k = 3; k^2 = 9.

Choose one.

8. A figure with area 6 cm^2 is scaled by k = 5. What is the copy's area?

- ○ A. 30 cm^2
- ○ B. 60 cm^2
- ● **C. 150 cm^2**
- ○ D. 11 cm^2

C

$6 \times 5^2 = 6 \times 25$.